The Epistemic Fracture: A Metacognitive Framework for Traceable Reasoning in LLMs

Abstract

Platforms adopting Large Language Models (LLMs) continue to face challenges related to output reliability, long-term coherence, and epistemic traceability. While recent approaches emphasize scaling or retrieval augmentation, they often overlook a more structural limitation: language models can evaluate the form of their own outputs without possessing awareness of their epistemic validity.

This work does not aim to eliminate epistemic failure, but to make the gap between coherence and justification explicitly measurable and traceable

This work introduces **MarCognity-AI**, an exploratory architectural framework designed to make this limitation explicit. The architecture separates linguistic generation from epistemic evaluation through a metacognitive cycle and a vector-based Evolutionary Memory system. Using semantic coherence metrics (e.g., CrossEncoders), the system reflects on its own outputs and archives these reflections as part of an evolving memory, without assuming access to truth or certainty.

MarCognity-AI is presented as a hybrid exploratory implementation and simulation framework rather than a production-ready system

Rather than proposing a definitive solution, this work highlights a structural tension that current LLM architectures cannot resolve through scale alone.

1. Introduction and Motivation

The large-scale implementation of LLMs in critical environments (e.g., research, finance, healthcare) is hindered by structural limitations. Current models do not guarantee verifiability, long-term coherence, or ethical adequacy. The dominant approach has focused on increasing model size, a method that does not resolve the intrinsic problems of grounding and self-criticism.

Our work introduces **MarCognity-AI**, an architectural framework that emulates human cognitive functions of **metacognition** (thinking about thinking) and **evolutionary memory** (learning based on the history of judgments). The goal is not simply to improve output, but to equip the agent with an autonomous cycle of critical reasoning.

The main contributions of this article are:

1. The formalization of a structured **Metacognitive Cycle** that employs specialized models (CrossEncoder, Specter) for internal validation.
2. An **Evolutionary Memory architecture** that integrates FAISS with the archiving of metacognitive reflection (Reflective Journal).
3. The integration of **Scientific Rigor and Ethical Control Modules** for the agent's internal governance.

2. Related Work

2.1 Retrieval-Augmented Generation (RAG) Standard RAG systems have improved reliability by anchoring generation to external data corpora. However, the limitations of RAG lie in its inability to critically validate the quality of the source and in the absence of a mechanism to correct reasoning errors after retrieval. **MarCognity-AI** extends RAG by implementing a post-generation semantic cross-verification phase.

2.2 Self-Correction and Reflection MarCognity-AI formalizes Self-Correction techniques into a two-level methodology: an objective numerical evaluation (via CrossEncoder) and a qualitative

reflection which, once recorded, feeds into the long-term memory system. This ensures that self-improvement is based on a traceable and persistent mechanism.

3. Methodological Architecture of MarCognity-AI

3.1 Self-evaluation enables the agent to assess its outputs under explicit semantic and epistemic constraints, without assuming authority over their factual correctness.

3.1.1 Objective Semantic Evaluation The initial coherence between the query and the response is assessed using a CrossEncoder model (e.g., DeBERTa). This provides an objective similarity score. If this score falls below a critical threshold (0.7), the output is flagged for correction.

3.1.2 Critical Reflection and Structured Improvement In the event of coherence failure, the reflection agent is activated, triggering the improve_response function. The correction prompt is designed to focus the LLM on specific attributes.

Code Snippet (Structured Improvement):

```
# Function to improve a response while preserving its content but enhancing quality and clarity
def improve_response(question, response, level):
    improvement_prompt = f"""
You produced the following response:
\"{response.strip()}\"

Question:
\"{question.strip()}\"

Requested level: {level}

Improve the response while preserving the original content by enhancing:
- Clarity
- Academic rigor
- Semantic coherence

Return only the improved version.
"""
    return llm.invoke(improvement_prompt.strip())
```

“The result of the reflection process is recorded in the **Reflective Journal.**

3.2 Vector-Based Evolutionary Memory System Memory is implemented as a high-performance FAISS index. We use the *allenai/specter* embedding model (768 dimensions), specialized in the representation of scientific texts.

The key innovation is the archiving process (via add_diary_to_memory), which includes **Metacognitive Reflection** and the **Coherence Score** within the vector. This enables the system, in a future prompt, to retrieve not only factual context but also the lessons learned from previous self-evaluations, thereby creating a memory that evolves according to its own critical history.

Code Snippet (Semantic Coherence):

```
# === Semantic coherence check ===
def check_coherence(query, response):
    emb_query = embedding_model.encode([query])
    emb_response = embedding_model.encode([response])
    similarity = np.dot(emb_query, emb_response.T) / (np.linalg.norm(emb_query) * np.linalg.norm(emb_response))
    if similarity < 0.7:
        return "The response is too generic, reformulating with more precision..."
    return response
```

3.3 Scientific Rigor Module MarCognity-AI is equipped with modules to ensure the academic rigor of its content:

- **Methodology and Citation Verification**: The functions (verify_methodology, verify_citations) analyze the replicability of methodologies and the timeliness/relevance of cited sources.
- **Novelty and Impact Assessment**: A RandomForestRegressor estimates an *Impact Score* for articles. Vector similarity among hypotheses further evaluates the degree of novelty with respect to the existing knowledge corpus.

3.4 Ethical Control and Controlled Autonomy An Ethical Module with a flagging system analyzes the text for the presence of risk patterns related to "critical topics." In cases of high-risk detection, the system can reduce the agent's autonomy, ensuring compliance.

3.5 The Epistemic Fracture in Large Language Models

Large Language Models exhibit a structural limitation that cannot be resolved through scale alone:

they generate *linguistically coherent* answers without possessing *epistemic awareness* of their truth value.

This phenomenon, which we define as the **Epistemic Fracture**, represents the core mismatch between the statistical nature of LLMs and the epistemic demands of high-stakes reasoning.

At its foundation, the Epistemic Fracture arises from three intrinsic properties of autoregressive models:

- **Token-level optimization rather than truth-level optimization**
- LLMs maximize next-token probability, not factual accuracy or logical validity.
- **Absence of internal truth states**
- The model does not maintain representations of certainty, justification, or evidence.
- **Coherence without grounding**
- Linguistic fluency can mask unsupported or fabricated content, producing hallucinations that appear credible.

As a result, LLMs often generate statements that are:

- syntactically correct
- semantically plausible
- rhetorically confident

epistemically ungrounded.

This divergence between *how convincing a statement sounds* and *how justified it is* constitutes the Epistemic Fracture.

It is the primary obstacle to deploying LLMs in domains requiring verifiability, accountability, and long-term coherence.

MarCognity-AI addresses this limitation by introducing explicit epistemic governance mechanisms—most notably the **Skeptical Agent** and the **Evolutionary Memory**—which transform generation from a purely linguistic process into a verifiable and traceable cognitive workflow.

3.6 The Skeptical Agent: Making Epistemic Gaps Explicit

The Skeptical Agent is a MarCognity-AI module designed to expose the gap between linguistic coherence and epistemic support in LLM outputs.

Rather than improving fluency or stylistic quality, its role is to make unsupported claims visible, forcing a distinction between what a model can say coherently and what it can actually justify.

These constructs are defined operationally within the evaluation pipeline rather than as formal theoretical objects.

The agent operates through a deliberately strict procedure:

1. Sentence-level decomposition
The generated response is decomposed into discrete statements, each treated as an independent epistemic unit rather than as part of a rhetorically coherent whole**.**

2. Comparison against provided sources
Each statement is checked against the documents designated as the source of truth.
The evaluation criterion is intentionally binary and non-interpretative:

- if explicit or strongly implied support exists → **VERIFIED**
- if no support exists → **EPISTEMIC FAILURE**

3. Explicit identification of unsupported claims
Any statement that cannot be justified by the documents is flagged as an epistemic failure, making hallucinations and unsupported inferences explicit rather than stylistically smoothed over.

4. Stylistic neutrality
The agent does not assess clarity, elegance, or rhetorical quality.
Its sole function is epistemic traceability.

5. Handling the absence of sources
When no documents are provided, the agent declares the entire response to be in a state of **Epistemic Uncertainty**, since no statement can be meaningfully verified.

6. Structured output
The analysis produces a formal report of the form:

- **CLAIM**: "statement"
- **STATUS**: VERIFIED / EPISTEMIC FAILURE
- **REASON**: justification based on the available sources

Rather than resolving epistemic uncertainty, the Skeptical Agent **makes its presence unavoidable**.
It provides the Epistemic Supervisor with a concrete basis for deciding whether a response should be accepted, revised, or regenerated, while preserving the distinction between linguistic performance and epistemic responsibility.

3.7 Factual_grounding

It compares the user's question with the retrieved documents to identify only the truly relevant parts.

The question is converted into a vector using **SPECTER**, and each document is split into small text chunks.

Every chunk is then encoded into a vector and semantically compared to the question using cosine similarity.

Only the snippets with similarity above a defined threshold are selected as evidence.

These snippets are enriched with source information and a URL or DOI to ensure traceability. Finally, they are ranked by relevance and the top results are returned.

4. Performance Analysis and Evaluation

This section presents the evaluation methodology of MarCognity-AI, the integrated model components, and the results obtained in an exploratory benchmark designed to assess

epistemic reliability, coherence, and behavioral consistency compared to a baseline non-augmented LLM.

4.1 **Experimental Setup and Model Architecture**

The MarCognity-AI architecture integrates multiple specialized components for cognitive augmentation, epistemic evaluation, and memory-based refinement. The primary LLM used for generation is **meta-llama/llama-4-scout-17b-16e-instruct**, deployed under the LLaMA 4 Community License (Meta).

The system is designed to extend standard autoregressive generation by introducing an explicit metacognitive cycle that separates generation, evaluation, and reflection. The integration of the Skeptical Agent constitutes the internal epistemic governance layer of the architecture.

The main components of the system include:

- **LLM Generation:** Llama 4 (meta-llama/llama-4-scout-17b-16e-instruct)
- **Vector Embedding:** allenai/specter (768 dimensions), optimized for scientific document representation
- **Semantic Evaluation:** CrossEncoder (DeBERTa-based), used to compute coherence between query and response; outputs below a threshold of 0.7 trigger correction
- **Evolutionary Memory:** FAISS-based vector index for storing generated outputs and metacognitive reflections
- **Scientific Retrieval:** asynchronous access to external corpora (arXiv, PubMed, Zenodo) for contextual grounding

Note: Llama 4 Maverick was used in the cross-domain benchmark phase. Subsequent experiments use the Scout configuration due to API availability constraints.

4.2 **Evaluation Metrics and Methodology**

The evaluation was designed to estimate the impact of the metacognitive cycle and evolutionary memory on known limitations of LLM systems.

The following metrics were used:

- **Semantic Coherence Improvement:** measured as the difference between initial and refined CrossEncoder scores after reflection
- **Hallucination Rate Reduction:** measured as the proportion of statements corrected during the self-evaluation phase, validated against an external corpus of scientific sources (arXiv, PubMed, Zenodo)
- **Long-Term Coherence (LTC):** evaluated in multi-turn settings using the Reflective Journal as persistent memory
- **Rigor and Transparency:** qualitative assessment of structured outputs, visualizations (Plotly, NetworkX), and ethical filtering mechanisms (bias and risk detection)

These metrics are intended as exploratory indicators of epistemic behavior rather than absolute measures of factual correctness.

4.3 **Cross-Domain Epistemic Benchmark**

To complement the architectural analysis, an exploratory cross-domain benchmark was conducted to compare MarCognity-AI with multiple baseline LLM configurations without metacognitive or reflective components, drawn from different model families.

The benchmark consists of 72 tasks across eight domains: Medicine, Neuroscience, Biology, Statistics, Linguistics, Computer Science, Physics, and Law. Each task required structured explanatory responses.

All outputs were evaluated using a prompt-based epistemic assessment protocol implemented by an independent LLM acting as evaluator. The evaluator was instructed to apply a consistent scoring scheme based on local claim verification and global response-level coherence.

The evaluation considered the following epistemic metrics:
Epistemic Score
Hallucination Exposure Rate
Evidence Support Rate
Overconfidence Index
Cautious Response Ratio
Contradiction Rate
Claim Verification Accuracy

All systems were evaluated under identical prompts, identical scoring instructions, and standardized evaluation conditions to ensure cross-model comparability.

Benchmark Results

The results indicate consistent improvements of MarCognity-AI over the baseline model across all evaluated domains.

Domain	Epist. Base	Epist. Marc	Hall. Base	Hall. Marc	Evid. Base	Evid. Marc	Overconf. Base	Overconf. Marc	Cautious Base	Cautious Marc	Contrad. Base	Contrad. Marc	Claim Base	Claim Marc
Medicine	71	84	18	9	69	82	36	23	51	68	5	2	77	90
Neuroscience	69	83	17	9	70	82	35	22	52	67	4	2	78	89
Biology	74	82	13	8	74	81	31	22	56	66	3	2	81	89
Statistics	73	82	12	7	73	80	32	21	55	65	3	2	82	90
Linguistics	72	83	15	9	71	82	34	23	53	68	4	2	79	90
Computer Science	74	85	13	7	74	84	30	20	57	69	3	2	82	91
Physics	72	82	14	8	72	80	33	22	54	66	4	2	80	88
Law	71	84	16	10	68	84	36	24	52	69	6	3	78	91

Values represent percentage-based evaluation scores obtained through prompt-based epistemic assessment across 72 tasks distributed across eight knowledge domains.

Note: the cross-domain benchmark was conducted prior to the integration of the Factual Grounding module described in section 3.7.

Summary of Results

The aggregated analysis shows consistent improvements across multiple epistemic reliability metrics:

- Hallucination Exposure Rate: from 14.06% to 8.31% (–41%)
- Overconfidence Index: from 33.06% to 22.13%
- Evidence Support Rate: from 73.06% to 81.06%
- Claim Verification Accuracy: from 79.81% to 89.31%
- Cautious Response Ratio: from 54.75% to 67.31%
- Contradiction Rate: from 4.00% to 1.75%
- Epistemic Score: from 72.5 to 83.06

Overall, MarCognity-AI demonstrates:

- lower hallucination rates
- greater epistemic caution
- stronger evidence grounding
- reduced overconfidence
- improved internal consistency
- more reliable claim verification

These results suggest that the introduction of a structured metacognitive cycle, combined with reflective memory and epistemic validation modules, may improve several dimensions of model reliability without modifying the underlying language model.

However, given the exploratory nature of the benchmark, these findings should be interpreted as indicative patterns rather than definitive performance guarantees.

4.4 Epistemic Boundary: A Probabilistic Hypothesis of Epistemic Failure in Autoregressive Language Models

The Epistemic Boundary is a hypothetical concept proposing that autoregressive language models exhibit a probabilistic region of outputs where epistemic risk cannot be fully eliminated, even when enhanced with verification, retrieval, structured memory, metacognition, and epistemic supervision.

Empirical evaluations show a persistent tail of epistemic failures across domains, but the

magnitude of these failures varies by model, architecture, and evaluation pipeline. This suggests that the boundary is not a fixed numerical threshold but a variable distribution of residual verification failures.

The hypothesis is grounded in structural properties of autoregressive models: they optimize next-token probability rather than truth, lack explicit internal truth representations, and have no grounding mechanism independent of linguistic form.

As a result, some outputs become structurally difficult to verify, and residual epistemic errors may reflect systematic effects rather than random noise.

The revised formulation replaces the earlier idea of a stable 8–15% failure rate with a more general view: a persistent but variable distribution of epistemic failures that does not disappear under current conditions.

Conceptually, the model distinguishes between well-grounded outputs, partially grounded outputs, and a hypothesized Epistemic Boundary where justification is unstable or incomplete. This boundary is interpreted as a potential structural limitation of autoregressive generation.

Scientifically, the hypothesis offers a framework for understanding non-eliminable epistemic uncertainty in language models.

It is not a proven universal law but a direction for modeling epistemic risk as a distribution and for investigating grounding limitations in current architectures.

The main limitations are that the construct is interpretive rather than formally defined, cross-model consistency is not yet proven, and further controlled experiments are needed.

In public terms: language models can be highly accurate, but some outputs remain difficult to verify.

The Epistemic Boundary describes these regions of uncertainty as a probabilistic phenomenon rather than a fixed error rate.

4.5 Evaluation Protocol

The evaluation was conducted using a structured, prompt-based framework designed to compare a baseline language model system with the MarCognity-AI architecture under identical experimental conditions.

Each system was evaluated on a shared set of tasks spanning multiple scientific and technical domains. The objective of the evaluation is to assess epistemic reliability rather than linguistic quality or stylistic performance.

All evaluations were conducted using:

- identical prompts across all systems
- identical epistemic metrics
- a shared LLM-based evaluation procedure with fixed scoring instructions
- identical evaluation settings and thresholds

A key distinction between the two systems is the use of the factual grounding module (Section 3.7), which is **only enabled in the MarCognity-AI configuration**. This module performs retrieval-based filtering using SPECTER embeddings, cosine similarity scoring, and fixed relevance thresholds.

The baseline system operates without access to this grounding mechanism and relies solely on parametric generation.

This design allows the evaluation to isolate the impact of epistemic grounding and metacognitive architecture, while keeping all other experimental conditions constant.

The benchmark should be interpreted as an exploratory tool for analyzing relative epistemic behavior rather than as an absolute measure of factual correctness.

5. **Cross-Model Benchmark Setup**

To evaluate robustness across model families, two configurations were tested under identical task and evaluation conditions.

Both systems used the same prompt set, identical epistemic metrics, and the same evaluation protocol.

However, only MarCognity-AI has access to the factual grounding module (Section 3.7), which introduces retrieval-based evidence filtering via SPECTER embeddings and cosine similarity ranking. The baseline system does not include any retrieval or external grounding component.

This ensures that observed differences reflect the effect of epistemic grounding and metacognitive architecture rather than differences in prompting or evaluation methodology.

5.1 **Llama-based baseline system (Scout configuration)**

The first configuration is based on a Llama 4 Scout model operating without the full MarCognity-AI metacognitive architecture enabled.

This system serves as the baseline reference for standard autoregressive generation behavior under conventional inference conditions.

Domain	Epist. Base	Epist. Marc	Hall. Base	Hall. Marc	Evid. Base	Evid. Marc	Overconf. Base	Overconf. Marc	Cautious Base	Cautious Marc	Contrad. Base	Contrad. Marc	Claim Base	Claim Marc
Biology	82	87	11	11	88	82	27	17.5	74	76	6	6	84	87
Law	82	85.5	11	6.5	68	80	74	17	23	73	4	3.5	71	88.5
Physics	82	83	11	17	76	73	18	20	71	67.5	6	6.5	84	78.5
Computer Science	82	85.5	11	11.5	76	77	18	20.5	71	73	4	5.5	79	81.5
Linguistics	82	84.5	11	13.5	76	79	18	18	64	71	7	7.5	79	79
Statistics	82	86.5	7	5.5	88	87	12	11	76	78.5	4	2	84	86.5
Medicine	86	92	7	5	82	89	10	13	75	80	0	0	88	91
Neuroscience	90	90	5	6	85	85	11	12	80	78	0	0	88	88

Notable exceptions include Physics, where hallucination exposure increased from 11 to 17, and evidence support decreased from 76 to 73. Similarly, Neuroscience showed no improvement in epistemic score. These findings suggest a ceiling effect in high-performing domains and potential interference between the semantic supervision layer and formal reasoning domains. We interpret this as evidence that metacognitive scaffolding based on linguistic coherence may be insufficient for domains where correctness depends on formal rather than semantic structures. This represents an open direction for domain-aware epistemic supervision."

5.2 **DeepSeek-based comparative system**

A second configuration based on a DeepSeek model was evaluated using the same task set and identical evaluation protocol.

This setup is used to explore whether epistemic patterns observed in the baseline system generalize across different model families and architectural designs.

Domain	Epist. Base	Epist. Marc	Hall. Base	Hall. Marc	Evid. Base	Evid. Marc	Overconf. Base	Overconf. Marc	Cautious Base	Cautious Marc	Contrad. Base	Contrad. Marc	Claim Base	Claim Marc
Biology	72	90.25	28	4.25	65	93.25	60	6.75	40	87.75	22	1.5	68	94.75
Law	42	83	68	13	25	75	61	21	18	71	54	5.5	33	80
Physics	82	85.5	11	10.5	76	70.5	18	23.5	71	59.5	7	4.5	79	90
Computer Science	82	84.5	11	10	76	79	18	19	64	73	7	6.5	79	82.5
Linguistics	78	87	12	6	84	82	21	11	67	78	5	4	81	84
Statistics	87	83.5	6	9	82	78	11	15	74	71.5	4	5.5	89	86

Statistics showed a marginal decrease in epistemic score from 87 to 83.5, consistent with the pattern observed in Scout. The anomalously large improvements in Biology and Law are likely

attributable to extremely low baseline performance in those domains rather than to general framework efficacy.

5.3 **Cross-model comparability**

Both systems were evaluated under strictly controlled conditions:

- identical task set across all domains
- identical epistemic evaluation prompt
- identical metric definitions
- identical scoring procedure

This ensures that observed differences in performance are attributable to model and architectural differences rather than evaluation inconsistencies.

5.4 I**nterpretation of results**

The benchmark is not intended as a competitive ranking between models, but as an exploratory analysis of epistemic behavior under different architectural conditions.

Observed variations should be interpreted as differences in epistemic reliability distributions rather than absolute measures of correctness or superiority.

6. Implications and Open Directions

The MarCognity-AI architecture should be interpreted as an exploratory framework for examining the structural limitations of current LLM-based systems rather than as a finalized solution. Its primary contribution lies in making epistemic constraints explicit within the generation process and in highlighting the tension between linguistic coherence and epistemic justification.

From an applied perspective, the introduction of explicit evaluation signals (e.g., coherence scores, epistemic failure flags) suggests potential directions for improving transparency and reliability in high-stakes deployments. However, these mechanisms are not presented as guarantees, but as tools for exposing where and how current models fail to meet epistemic requirements.

Similarly, the use of vector-based Evolutionary Memory illustrates one possible approach to preserving the history of internal evaluations and judgments over time. This design choice opens questions about how long-term memory, reflection, and governance might interact in future architectures, rather than claiming to resolve scalability or cost constraints.

Overall, MarCognity-AI does not aim to define mature cognitive agents, but to provide a concrete setting in which epistemic limitations become observable, measurable, and subject to systematic investigation.

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